# boundline

## Borrow against your stocks on Base. Paid in your currency.
## Never liquidated while Wall Street is closed.

The session-aware credit layer for Coinbase's tokenized stocks on Base: an onchain market calendar, oracles that refuse a frozen price, protection that runs on weekends, and credit lines any wallet or broker can embed. Non-custodial.

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Farseen Shaikh · TODO: second developer · Colosseum Crypto World's Fair, Base track · October 2026

# The stocks trade all weekend. Their price does not.

| Feed | Held print | Age at reopen | Rounds while closed | First print | Gap |
|---|---|---|---|---|---|
| AAPLc | 336.84 | 52.5 h | 0 | +51 s | –5.8 bps |
| AMZNc | 261.80 | 52.7 h | 0 | +41 s | +51.9 bps |
| GOOGLc | 352.63 | 54.7 h | 0 | +31 s | +4.7 bps |
| METAc | 719.92 | 54.4 h | 0 | +53 s | +14.7 bps |
| MSFTc | 536.08 | 53.3 h | 0 | +23 s | –1.5 bps |
| MSTRc | 154.61 | 53.2 h | 0 | +21 s | +201.0 bps |
| NVDAc | 230.26 | 56.7 h | 0 | +23 s | +1.1 bps |
| SNDKc | 1,581.77 | 52.4 h | 0 | +21 s | +144.7 bps |
| SPCXc | 162.62 | 52.1 h | 0 | +37 s | +94.9 bps |
| TSLAc | 383.97 | 55.9 h | 0 | +23 s | +16.5 bps |

- The Chainlink feeds for Coinbase's tokenized stocks publish **Sun 20:00 → Fri 20:00 ET** and hold Friday's close in between.
- Aave V4′s Equities Hub and Morpho's B20 markets read that held price with **no staleness, pause or market-hours check**. Liquidation is open at any hour.
- At the reopen the weekend's move prints in **one block**. A position that gapped through its limit is gone before its owner can act.

Base's docs: "never settle or liquidate against a frozen feed". Aave's risk provider calls session-aware handling "a future stage".

The 9–11 Oct 2026 weekend, all ten feeds: captured live by the Observatory and reconstructed from the Chainlink round history on Base mainnet, agreeing on every field (`docs/evidence/reopen-2026-10-11.md`). 2–4 Oct looked the same: zero rounds, first prints 23–53 s, MSTRc +282 bps.

WHAT IT IS

# A credit layer that knows when Wall Street is closed.

## Refuse

a frozen price at the venue: on Boundline's markets nobody can borrow or liquidate against it; repay and add-collateral always work.

## Protect

on Saturday, at the price the market actually shows: a small sale paid for by the collateral, or a repay from a reserve.

## Embed

a line of credit any wallet or broker can offer, paid out in the user's own currency, with one record per loan.

- **Who pays:** wallets, neobrokers and fintechs that list tokenized stocks (B2B2C); curators who want protected markets; issuers who need credit rails.
- **Non-custodial:** the user keeps their own Morpho or Aave position and authorises Boundline, or the venue's own audited contract, with one signature.

HOW IT IS BUILT

# From the clock to the partner: every layer is live.

| Layer | What decides | Proof |
|---|---|---|
| Time | `MarketCalendar`: every NYSE session onchain (pre, regular, post, overnight, weekend, holiday, early closes, daylight saving) | 5,643 instants, 3 independent implementations, 0 disagreements |
| Verdict | `SessionGate`: per stock, is this price safe? Session, feed age, Coinbase OracleRegistry pause, issuer transfer pause; fails closed | 34 oracle tests |
| Venue oracle | `SessionGuard` (Morpho): reverts while the feed is held, so borrow, withdraw and liquidate stop; repay never reads it | live since 3 Oct; mainnet fork |
| Protection | `ShadowGuard`: the 24/7 Aerodrome TWAP less a haircut. Morpho's audited pre-liquidation + Boundline's `Protector`; the `DeskProtectionVault` ladder and the `BoundlineReserve` (ERC-4626) | keeper acted on its own, also under a frozen feed; Reserve advances settled |
| Credit | `CreditDesk` (Morpho), `AaveDesk` (Aave V4 via Taker/Giver): partner policy, decision hashes, local-money payout | live lines; Aave on a mainnet fork |
| Data | Observatory, indexer with hash-chained daily roots, fork replay through weekend gaps | live on mainnet reads |
| Partners | API (x402-metered), SDK, app, console, MCP agent surface | eight Workers live |

ONE WEEKEND, PROTECTED

# Repay is always open. Sales wait for the session.

**Fri 20:00 ET**
The feed freezes. `SessionGuard` stops pricing it: no borrows, no liquidations on Boundline's markets. Repay and add-collateral stay open.

**Saturday**
The token moves on Aerodrome. `ShadowGuard` prices the position at the TWAP less 2%. The keeper sizes the action **on chain**.

**Saturday, inside the band**
A slice of collateral is sold inside Morpho's pre-liquidation callback and repays the debt; or the vault repays from the user's reserve, then a bounded Reserve advance.

**Sun 20:00 ET**
The feed reopens. The position is already safe. Any advance is settled in session, at a fresh price, with its fee.

**Why this order is forced:** Morpho's pre-liquidation refuses once a position is past its limit, so nothing can rescue a position that gaps through it at the reopen. Only an action taken **while the feed is frozen**, at the live price, prevents the liquidation. The venues cannot see the live price; Boundline does.

# Base itself holds the keeper's action until the feed state matches.

- Each protective action is signed with a **storage predicate** on the feed's `updatedAt` slot and a block bound, sent with `base_sendRawTransactionValidity`.
- "**Only while held**" → a Saturday action can never land after the feed has printed a new price.
- "**Only once it prints**" → the first-block action waits in the sequencer for the reopen.
- The sequencer accepts a bound at most 32 blocks ahead, so the keeper re-arms every minute; every check also lives in the contract.

| Predicate | On Base Sepolia |
|---|---|
| `slot = held` | `0x6b17288f…49dd` · included next block |
| `slot ≠ held` | `0xbbd84631…b2fb` · included after the round changed |
| keeper, held | `0x982c7327…b695` · automatic protection |
| keeper, held | `0x6d45ddc9…40d9` · automatic, 89 min later |
| keeper, frozen | `0x9774913e…0352` · close-out, staged outage, 12 Oct |
| vault, frozen | `0x47ac5279…1d24` · ladder repay, same predicate |

Measured: "not before", not "as soon as" (`docs/evidence/validity-transactions-sepolia.md`).

WHAT IS LIVE

# Base Sepolia, on real B20 test stocks mirrored from mainnet.

| What | What happened | Transaction |
|---|---|---|
| Weekend refusal | Sat 3 Oct: collateral added; a borrow attempt reverts `MarketClosed(WEEKEND)` | `0x01191a77…760b` |
| In-session draw | Tue 6 Oct, PRE: 5 USDC through the CreditDesk at $332.67, decision hash on chain | `0x661ebb1a…8bac` |
| Automatic protection | keeper pre-liquidated an in-band position at the live price: shadow LTV 60.12% → 58.00% | `0x982c7327…b695` |
| Reserve advance | protective repay on a live line: 0.5 USDC from the user's reserve + 0.2 USDC Reserve advance, settled with a 1% fee | `0x28c1ec44…08a4` |
| Local money | 0.5 USDC → 0.46 EURCt inside the borrow flow, session-checked | `0x9090633c…5bd3` |
| x402 | two keyless API calls paid $0.01 test USDC each | `0x8816d08f…c6ff` |
| Frozen feed | Mon 12 Oct, a staged feed outage: the keeper closed out an in-band position at the live price as a validity transaction; a raw liquidation reverted `FeedStale`; the vault repaid a second line from the user's reserve + a Reserve advance | `0x9774913e…0352` |
| Drills | stale feed, registry pause, issuer pause, scheduled multiplier, blocked receiver, keeper hold: each refused by name, repays still land | `0x84064c22…8429` |
| Webhooks | a hold and a release on a live line reached the partner HMAC-signed, 84 s and 170 s after the block | `0x5dd9d8df…42b7` |
| Dividend → loan | a dividend multiplier step swept into a desk line's debt: flash loan, repay first, sell only the dividend collateral | `0xa44b7867…f680` |

Every Sepolia contract verified on sepolia.basescan.org (`deployments/base-sepolia.json`). The same footprint is on Base mainnet since 13 Oct: 50 transactions, 16 contracts verified on basescan.org, 33 of 33 checks (`deployments/base-mainnet.json`); its market is unseeded and nobody has opted in yet.

BASE MAINNET

# Deployed 13 Oct; read live every minute, proven on forks, replayed through weekend gaps.

**$1.73M**

USDC borrowed on Aave V4's MAG7 stock spoke against $5.14M of tokenized-stock collateral (block 52,517,725, 12 Oct)

**14**

Morpho B20 markets, 16,315.76 USDC borrowed (12 Oct)

**10**

Coinbase stock feeds, sessions and registry flags, every minute

**8**

mainnet-fork suites: SessionGuard on the live Morpho, the Protector on a real Steakhouse borrower, Aave's Taker and Giver, the EURC pool, the replay, the Aave hub halt, a bad-debt probe, a real dividend

**Replay, 7 Oct (fork block 52,295,895)**

| Gap at reopen | Raw venue: liquidated | Boundline: liquidated |
|---|---|---|
| –3% | 0 of 5 | 0 of 5 |
| –8% | 1 of 5 | 0 of 5 |
| –15% | 2 of 5 | 0 of 5 |

Five positions: three as they stood, two real positions staged near their limit. Every raw liquidation was a staged position; the three as they stood stayed healthy even at –15%. Boundline acts on Saturday at the live price (pre-liquidation in the band, a repay above it). Evidence hash `0xa626f099…3c45`.

THE PARTNER SURFACE

# A wallet integrates in a day and keeps its brand.

**Partner API**
Keys, quotes (capacity rounds down, charges round up, unknown data fails closed), obligations, decision timelines, policy versions, drills, evidence export. Public routes metered by **x402**. Route plans; HMAC-signed webhooks.

**SDK + app**
One `wallet_sendCalls` batch: authorise, approve, add collateral, draw, swap to local money. Preconfirmation in 200 ms. The Boundline app is partner #1.

**Console + agents**
Obligations, timelines, policy editor, drills, reconciliation, evidence with the hash recomputed in the page. An **MCP** server: session, positions, shadow health, quote, protect, route.

Live: app · console · API · agent

PUBLIC GOODS

# MIT, deployed, usable by anyone without asking us.

| Public good | What anyone can do with it |
|---|---|
| B20 Credit Observatory | read every Coinbase stock feed, session, registry flag, DEX gap and real Aave and Morpho position on Base, minute by minute, from an open JSON API |
| `MarketCalendar` + `@boundline/calendar` | ask onchain or in TypeScript which NYSE session any instant belongs to, holidays and early closes included |
| `SessionGate`, `SessionGuard`, `ShadowGuard` | list a Morpho market that refuses a frozen price; protect positions at a live price |
| The weekend dataset | two weekends recorded (9–11 Oct also captured live); rebuild any weekend's freeze and reopen from the chain (`scripts/reopen-evidence.py`, standard library only) |
| The replay harness | take real positions through weekend gaps on a fork and publish the result with a hash (first run 7 Oct; scheduled nightly in CI once the repository is public) |

# Fees on credit that already exists on Base.

## MODEL

Platform fee per partner · 20–50 bps on originations · 10–25 bps a year on protected notional · x402-metered API · Reserve advance fees and the pre-liquidation incentive to capital providers.

## TEAM

Farseen Shaikh, incoming MSc Data Science, TU Berlin · TODO: second developer, one line each

118 commits from 3 to 13 Oct, 264 Foundry + 898 TypeScript tests, eight mainnet-fork suites, eight services live, the footprint on Base mainnet.

## THE ASK

### $1M seed, 12 months

40% engineering · 20% legal and structure · 15% security (two audits) · 15% capital and partnerships (seed the Reserve) · 10% go-to-market

## MILESTONES

Three design partners live on mainnet · $5M protected notional · the oracle adopted by a curator · a third venue

## VALIDATION

$1.73M of stock-backed debt on Aave V4’s stock spoke alone sits on a feed that freezes every weekend.

TODO: conversations, opt-ins



# boundline

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| | |
|---|---|
| **Observatory** | https://observatory.boundline.xyz |
| **App** | https://app.boundline.xyz |
| **Console** | https://console.boundline.xyz |
| **Site** | https://boundline.xyz |
| **Code** | github.com/FarseenSh/boundline · MIT |
| **Every claim** | `docs/submission/SUBMISSION.md` ⟶ evidence index; `deployments/base-sepolia.json` |